UX Research Questionnaire & Unsupervised Clustering System Design

Architecture and mathematical specifications for classifying insurance agents by empirical behavioural patterns and operational workstyles rather than static demographic rules.

Instead of hardcoding a predefined target set of personas into the portal, the architecture evaluates empirical response distributions. A downstream unsupervised clustering pipeline identifies natural agent cohorts in count, shape, and behavioural character.

1. Operational Vector Space Framework

Every respondent is mapped into a **14-dimensional coordinate vector**:

$$R = [Skill_1, Skill_2, Skill_3, Skill_4, Skill_5, Journey_1, Journey_2, Journey_3, Journey_4, Journey_5, Journey_6, WP, AR, PR]$$

The 14 standardized dimensions capture three complementary behavioural facets:

1. **Five Core Skills (Q7)**: Device / System Familiarity, Insurance Knowledge, Sales Pitch & Communication, Customer Relations & Service, and Digital Tools & Systems.
2. **Six Journey Task-Satisfaction Ratings (Q27)**: Task 1 (Scheduling), Task 2 (Policy Sales), Task 3 (Underwriting), Task 4 (Policy Servicing), Task 5 (Branch Admin), and Task 6 (Training & Learning).
3. **Three Workstyle & Behavioural Indices**: Workaround Propensity (WP), Abandonment Rate (AR), and Peer Reliance (PR).

> **System Utility Perception (SUP)**: SUP reflects overall perceived efficiency and tool reliability (derived from Q10 and Q11). SUP is calculated and displayed on persona overview cards and radar diagnostics, but **is not an input to the clustering vector**.

Demographic & Tenure Variables (Age, Current Job Title/Position, Years of Experience in Insurance) are deliberately excluded from the clustering vector `R`. They are retained strictly as post-hoc descriptive variables to evaluate across discovered clusters (e.g. *"does this cluster skew toward junior agents?"*), never as inputs that define or constrain cluster boundaries.

2. Vector Formulas & Questionnaire Data Sources

Dimension	Code	Role	What It Captures & Data Source
System Utility Perception	`SUP`	Display / Diagnostic	Perceived efficiency and utility impact of official company digital systems and tools (Q10, Q11).
Workaround Propensity	`WP`	Clustering Input	Propensity to use non-standard personal tools (LINE, Google Calendar, paper) to bypass official systems, and stated reasons (Q13, Q14, Q23).
Abandonment Rate	`AR`	Clustering Input	Frequency of system aborts, process dropouts, repeated data entry, and manual fallbacks across all six journey tasks (Task 1–6 Pain Points).
Peer Reliance	`PR`	Clustering Input	Reliance on senior colleagues, branch assistants, supervisors, or peer networks to complete operational tasks (Task 1–6 Tools & People, Q23).

Detailed Index Formulas

Workaround Propensity (WP)
Calculates reliance on non-official tools and methods: `WP = min(5.0, 1.0 + (Q13_reasons × 0.5) + (Q14_methods × 0.3) + (Q23_tools × 0.4))`
Parsed using punctuation delimiters (`/[;； ,，]+/`) without whitespace splitting, ensuring multi-word terms like *Google Calendar* are counted as single entries. Company systems (Agent Portal, Mobile App, AP System, ABC Academy) are excluded from the workaround tally.

Abandonment Rate (AR)
Evaluated via keyword/semantic tracking of system dropouts, process aborts, and manual fallbacks across all 6 journey tasks. In synthetic and live surveys, general pain points are common across tasks; AR specifically identifies operational failure states where the agent gives up on official digital tools.

- **Documented English Dropout Keywords**: `"by hand"`, `"converting pdf"`, `"re-entering"`, `"re-enter"`, `"twice"`, `"rejected"`, `"re-upload"`, `"re-uploading"`, `"password reset"`, `"chasing a colleague"`, `"logging back in"`, `"system error"`, `"give up"`, `"giving up"`, `"abandon"`, `"manual"`
- **Documented Traditional Chinese Dropout Keywords**: `"手動"`, `"手寫"`, `"重複輸入"`, `"重填"`, `"退件"`, `"重新上傳"`, `"重設密碼"`, `"密碼"`, `"重新登入"`, `"強制登出"`, `"被登出"`, `"系統錯誤"`, `"錯誤"`, `"放棄"`
- **Linear Scaling Formula**: `AR = 1.0 + (matched_dropout_tasks / 6.0) × 4.0` (producing a continuous, calibrated score from 1.0 to 5.0).

Peer Reliance (PR)
Detects operational dependency on colleagues, branch assistants, or mentors across all 6 journey tasks and personal contact channels (Q23). Matches keywords such as *senior colleague* (資深同事), *administrative assistant* (行政助理), *peer support* (同仁協助 / 同儕支援), *supervisor* (主管), and *mentor* (前輩):
`PR = min(5.0, 1.0 + peer_signals × 0.5)`

3. Downstream Unsupervised Cluster Discovery Logic

The clustering engine analyses agent vectors without structural preconceptions, pre-assigned centroids, or static archetype quotas. The discovery pipeline proceeds through four sequential stages:

1. **Coordinate Normalization**: The 14-dimensional raw vector is standardized using z-score normalization (`(x − mean) / std`) so that differing measurement scales do not distort distance calculations.
2. **DBSCAN Outlier Pre-filtering**: Density-based spatial clustering identifies isolated respondents and extreme response patterns (labelled as `outlier` with `_dynamicClusterId = −1`), protecting cluster centroids from noise.
3. **Dynamic Cluster Identification**:

   > K-means groups the respondents who remain after DBSCAN outlier filtering. The analyst chooses K (3–8, default 5) in the dashboard; outliers are shown as a separate group. Silhouette-based K suggestion is not yet implemented.

   The number of discovered clusters is not fixed to any predefined count but varies **between datasets** to reflect true empirical differences across surveyed cohorts.
4. **Archetype Synthesis & Labelling**: Clusters are ranked by average digital utility, and distinctive dimensions (z-score extremes) are extracted live to construct descriptive persona profiles.

4. Implications for Dashboard Implementation

Because the portal is completely data-driven, the dashboard must treat cluster count, centroid metrics, and qualitative summaries as dynamic outputs rather than static assets:

- **Variable Tab Navigation**: Supports variable cluster tabs (K = 3 to 8) plus an optional Outlier tab, populated dynamically from the active clustering result rather than expecting 5 fixed personas.
- **Live Dynamic Archetype Ordering**: Discovered clusters are mapped dynamically (`c0`, `c1`, `c2`...) and ordered logically by digital adoption trajectory.
- **Dynamic Persona Profiles**: Core traits, journey timelines, emotional arcs, and top characteristics are generated directly from empirical cluster averages.
- **Demographic Cross-Tabulations**: Demographic cards (Location, Rank, Age, Client Scale) display the empirical composition of the selected cluster post-hoc.
- **Editable Synthesis Step**: Persona names, quotes, and qualitative summaries are fully editable in the UI, empowering researchers to save nuanced interpretations directly on top of the mathematical clusters.

> **Operational Summary**: The portal decouples objective mathematical grouping (14-dimensional DBSCAN + K-means clustering) from subjective qualitative interpretation (human persona synthesis). This ensures personas reflect authentic user needs and operational friction rather than preconceived archetypal biases.